

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

26AI5707: Pattern Recognition and Machine Learning

Report1: Familiarizing with Matrices, and Distributions

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Experiment No.	1
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System No.	-

OBJECTIVE:

To implement basic matrix operations and generate Gaussian random variables, thereby understanding the mathematical concepts behind these operations.

MATRIX MULTIPLICATION:

Procedure:

1. Checking whether the matrices A , B are in the form $m \times n$ and $n \times p$ orders respectively ($\therefore$ the solution matrix S will be in the form $m \times p$)
2. Multiplication of rows of A with columns of B
3. The product of i^{th} row of A and j^{th} row of B is placed in S_{ij} .

DOT PRODUCT

Procedure:

1. Checking whether the vectors V_1 and V_2 have the same size
2. Multiplying corresponding elements of the vectors V_1 and V_2
3. Summation of all the products

TRANSPOSE:

Steps:

For every $(i,j)^{\text{th}}$ element of matrix A , place them in $(j,i)^{\text{th}}$ position in A'

TRIANGULAR MATRIX

Steps:

1. Take transpose of A i.e., A'
2. If $A' = A$, Then the matrix is transpose

SYMMETRIC MATRIX:

Steps:

1. For $(i,j)^{\text{th}}$ element of matrix A :
 - a. Check if $i > j$ (for Upper triangular matrix)
(or) $i < j$ (for Lower triangular matrix)
 - b. If true, check if the value is 0

2. If all elements in the condition are zero, then the matrix is Triangular Matrix

UNIFORM AND GAUSSIAN RANDOM VARIABLE:

Steps:

1. Generate two independent uniform random variables U_1 and U_2 where $U_1, U_2 \sim \text{Uniform}(0, 1)$
2. Calculate $s = U_1^2 + U_2^2$
3. Compute $k = \sqrt{\frac{2 \ln s}{s}}$
4. The resulting k follows the standard normal distribution:

$$k \sim \mathcal{N}(0, 1)$$

For a Gaussian with mean μ and standard deviation σ :

$$x' = \mu + \sigma x$$

$$y' = \mu + \sigma y$$

OVERALL LEARNING:

This laboratory helped me understand the implementation of fundamental matrix operations and random variable generation without relying on NumPy. It strengthened my programming skills and provided a foundation for Pattern Recognition and Machine Learning concepts.

CONCLUSION:

The objectives of the experiment were successfully achieved. All required matrix operations were implemented correctly, and Gaussian random variables were generated. The output histograms confirmed the expected behavior of uniform and Gaussian distributions.